Qiyan Jin

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Research Interests

My research interests lie in developing models and learning algorithms for embodied AI and robot learning, especially methods that generalize across tasks and environments. Directions of interest include multimodal foundation models, vision-language-action models, world models, reinforcement learning, and imitation learning. I am also open to emerging approaches to representation learning, decision-making, and robot adaptation.

Education

Nanyang Technological University, Singapore Aug 2026 – Present

School of Electrical and Electronic Engineering

Master of Science (Computer Control & Automation) · Full-time coursework
Expected programme completion: May 2027

Shanghai Jiao Tong University Sep 2022 – Aug 2026

University of Michigan–Shanghai Jiao Tong University Joint Institute

Bachelor of Science in Engineering, Mechanical Engineering · GPA: 3.0/4.0
Undergraduate exchange: Nanyang Technological University, Singapore · 2025–2026 academic year

Research Experience

DexMate Oct 2026 – Present

Robot Learning Intern · Singapore

- Investigating **3D Gaussian Splatting (3DGS) digital twins** for sim-to-real robot learning.
- Developed a **composable task-evaluation framework**, combining reusable geometric and contact predicates with temporal stage graphs for manipulation progress tracking and failure diagnosis.
- Validated the prototype using recorded simulation trajectories and end-to-end smoke tests, with physical measurement traces and diagnostic reports for reproducible review.

Shanghai Institute for Physical Artificial Intelligence and Robotics (PAIR) May 2026 – Oct 2026

Research Intern · DexArena: π0.5 Adaptation for Dexterous Manipulation

- Adapted pretrained **π0.5** for DexArena, a benchmark for multi-embodiment, multi-stage dexterous manipulation. Extended action projections for **29-DoF arm–hand and 58-DoF bimanual control**, and fine-tuned both the vision-language backbone and action expert with LoRA.
- Developed stage-pooled training for **10 long-horizon task families / 30 subtasks**, using joint-wise normalization and masked flow-matching supervision for heterogeneous demonstrations. Used one policy per long task, conditioned on environment-provided stage instructions.
- Implemented **latency-aware asynchronous action-chunk fusion**: aligned predictions to observation timestamps, discarded stale actions, and blended overlapping chunks in physical joint space to maintain continuous execution.
- Compared full-parameter and LoRA fine-tuning; adopted LoRA for its much lower training cost, at slightly lower success. Controlled comparisons showed improved subtask success with asynchronous fusion.

Co-author of the associated DexArena manuscript (ICLR 2027, under review); responsible for its π0.5 baseline.

Sharpa Wave Cross-Simulator Evaluation Aug 2026 – Oct 2026

Independent Research Project · MuJoCo and kit-less OVPhysX

- Built controlled cross-simulator tests for publicly available dexterous-hand models, aligning joints, scenes, and control inputs. Examined motion, fingertip trajectories, contact, and object retention through repeated runs and timestep refinement.
- Stabilized contact readouts by changing stepping calls. Dry-friction controls explained most fingertip differences in the tested non-contact tasks; remaining causes were not fully isolated.
- Used collision-disabled controls to confirm that free-ball retention depended on hand contact. Both backends retained the ball, but successful retention did not establish matching trajectories; some OVPhysX cases remained timestep-sensitive.

Manuscript: "Hidden Simulator Choices in Dexterous Hand Evaluation: A MuJoCo–OVPhysX Diagnostic Case Study." Independent manuscript, 2026.

Research Projects

Unitree G1 Whole-Body VLA Simulation May 2026 – Aug 2026

Undergraduate Capstone · Five-person team

- The team explored humanoid whole-body manipulation in MuJoCo, using teleoperated demonstrations to train GR00T N1.7 policies and SONIC WBC for whole-body control.
- Primarily handled PICO 4 / VR_3PT whole-body teleoperation. Helped calibrate the headset, controllers, and simulated robot; collected walking, grasping, and bimanual-handover demonstrations.
- Reviewed task labels, episode boundaries, and action completeness in long-horizon demonstrations; identified control anomalies and recollected low-quality trajectories. Reproduced control degradation during video recording, contributed to software and workflow changes, and rechecked teleoperation with recording enabled.

Teaching Experience

Teaching Assistant · ME3950J Experiment (1) Fall 2025

Technical Communication component · Shanghai Jiao Tong University

- Prepared approximately 3–4 hours of recorded tutorials explaining report templates and grading rubrics. Reviewed report formatting, helped assess presentations and Design Reviews, and compiled end-of-semester grades.

Teaching Assistant · ME2500J Design and Manufacturing I Spring & Summer 2025

Shanghai Jiao Tong University · Two course offerings

- Led weekly 2–3-hour laboratory sessions covering SolidWorks CAD, Arduino, and project development. Advised students on mechanical design and implementation; helped grade assignments, exams, Design Reviews, and final presentations.
- Worked with the instructor to develop project objectives, vehicle and bridge specifications, Game Day rules, and scoring criteria. Helped revise the second offering's challenge to require a U-turn and a second, narrower bridge.

Selected Engineering Experience

Dual-Arm Dental Navigation Simulation Junior Fall

ME4670J Introduction to Robotics

Derived a six-frame coordinate-transform chain between surgical and vision arms for an occluded target. Used Python and MATLAB to test inverse kinematics, configuration space, and RRT* path planning.

Automatic Card Dealer for Multiple Players Sophomore Summer

ME2500J Design and Manufacturing I

Designed the dealing mechanism and top-mounted electronics to rotate together, reducing cable winding. Adjusted card-release timing for rotational speed to enable continuous delivery to different positions.

Ronovo Surgical · Project Assistant Intern Feb 2025 – Apr 2025

Supported medical-robotics scheduling and R&D milestone tracking across hardware, software testing, and supply chain teams. Organized technical documents and bills of materials; assisted with procurement, prototype assembly, and basic testing.

Technical Skills

Programming & Learning: Python, MATLAB, PyTorch; LeRobot, π0.5 SFT
Simulation & Evaluation: ManiSkill, MuJoCo, kit-less OVPhysX; task development, policy evaluation
Teleoperation & Data: Calibration, demonstration collection, and quality review
Design & Fabrication: SolidWorks/CAD, 3D printing, laser cutting, prototype assembly and testing

Selected Honors

First Prize, Shanghai Korean-Chinese University Student Scholarship · Mar 2024
Third Prize, SJTU "Liming Cup" Mechanical Creativity Competition · Apr 2023
First Prize, ME2500J Game Day

Community Outreach

Bilingual Educational Outreach Speaker July 2022

Yanbian, China

- Invited by the Yanbian prefectural government to deliver five talks in Mandarin and Korean to approximately 400–500 high school students, introducing university study in computer science and mechanical engineering.
- Provided follow-up guidance on university and major selection, and encouragement during students' preparation for the national college entrance examination.